

```

>
>
>
> restart;
> with(Statistics) :
>
> with(stats):
> with(stats[statplots]):
>
> #Poisson distribution
>  $P_m := \frac{\lambda^m}{m!} \cdot \exp(-\lambda);$ 
>
>  $P_m := \frac{\lambda^m e^{-\lambda}}{m!}$  (1)
>  $\lambda := 20;$ 
>  $\lambda := 20$  (2)
>  $n := 40;$ 
>  $n := 40$  (3)
> for m from 0 to n do
>    $P[m] := \text{evalf}(P_m);$ 
>    $dd[m+1] := \text{Weight}(m..m+1, P[m]);$ 
> end do;
>
> for m from 0 to n do
>   printf("%f %f \n", m, evalf(P[m])) :
> end do;
0.000000 0.000000
1.000000 0.000000
2.000000 0.000000
3.000000 0.000003
4.000000 0.000014
5.000000 0.000055
6.000000 0.000183
7.000000 0.000523
8.000000 0.001309
9.000000 0.002908
10.000000 0.005816
11.000000 0.010575
12.000000 0.017625
13.000000 0.027116
14.000000 0.038737
15.000000 0.051649
16.000000 0.064561
17.000000 0.075954
18.000000 0.084394
19.000000 0.088835
20.000000 0.088835
21.000000 0.084605

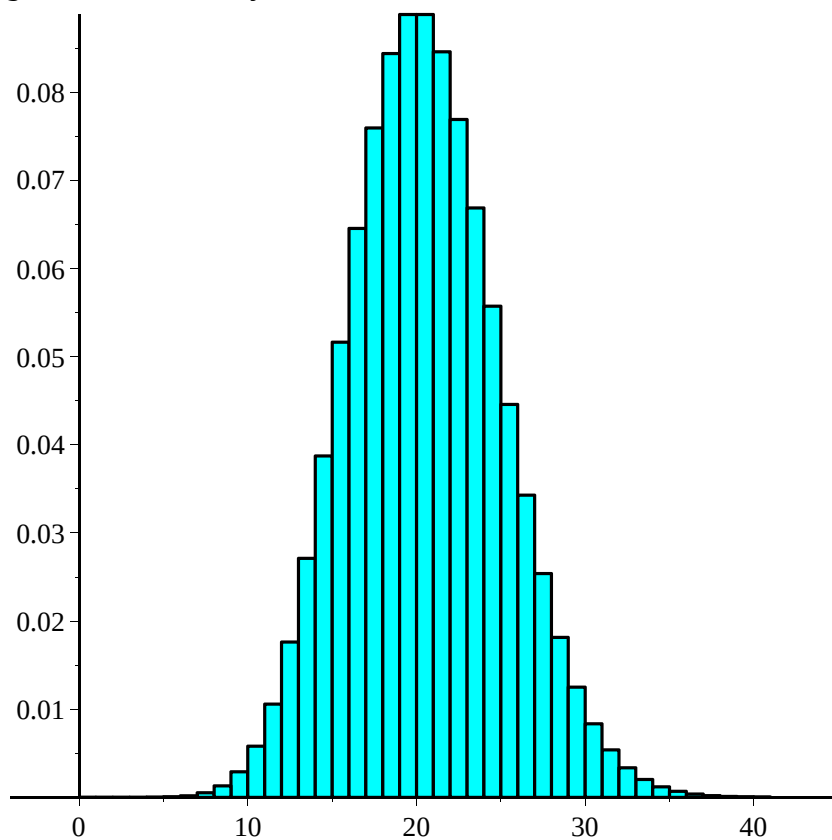
```

```
22.000000 0.076914
23.000000 0.066881
24.000000 0.055735
25.000000 0.044588
26.000000 0.034298
27.000000 0.025406
28.000000 0.018147
29.000000 0.012515
30.000000 0.008344
31.000000 0.005383
32.000000 0.003364
33.000000 0.002039
34.000000 0.001199
35.000000 0.000685
36.000000 0.000381
37.000000 0.000206
38.000000 0.000108
39.000000 0.000056
40.000000 0.000028
```

```
>
```

```
> dP := [seq(dd[i], i = 1..n + 1)]:
```

```
> histogram(dP, color = cyan);
```



<pre> > $\mu_x := 0;$ = > for m from 0 to n do $\mu_x := \mu_x + m \cdot P[m] :$ end do: = > $\text{print}(\mu_x=, \mu_x);$ = </pre>	<pre> $\mu_x := 0$ </pre>	<pre> (4) </pre>
<pre> = > $D_x := 0;$ = </pre>	<pre> $\mu_x=, 19.99893596$ </pre>	<pre> (5) </pre>
<pre> = > $D_x := 0;$ = </pre>	<pre> $D_x := 0$ </pre>	<pre> (6) </pre>
<pre> > for m from 0 to n do $D_x := D_x + (m - \mu_x)^2 \cdot P[m] :$ end do: = > $\text{print}(\text{`D}_x=`, D_x);$ = </pre>	<pre> $D_x=, 19.98782560$ </pre>	<pre> (7) </pre>
<pre> > $\sigma := \text{sqrt}(D_x);$ = </pre>	<pre> $\sigma := 4.470774608$ </pre>	<pre> (8) </pre>
<pre> = > = > = > = > </pre>		